Lesson 3.13 - How to Calculate Variance

> **Video:** How To Calculate Variance by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Formula (sample variance)

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

The variance is the **square of the standard deviation**, so it is the same calculation without the final square root.

Example 1

Data: 6, 9, 14, 10, 5, 8, 11 → sorted: 5, 6, 8, 9, 10, 11, 14 ($n = 7$)

Mean: $\bar{x} = \frac{\sum x}{n} = \frac{63}{7} = 9$

x	$x - \bar{x}$	$(x - \bar{x})^2$
5	−4	16
6	−3	9
8	−1	1
9	0	0
10	1	1
11	2	4
14	5	25
Sum	0	**56**

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{56}{7 - 1} = \frac{56}{6} \approx 9.33$$

What variance tells you

Variance measures **spread**: how far the data lies from the mean. A **larger variance means the data is more spread out**.

Example 2 - same mean, different spread

Data set 1	Data set 2
6, 7, 8, 9, 10	4, 6, 8, 10, 12
mean = 40/5 = **8**	mean = 40/5 = **8**

On a number line, both sets are centered at 8. Set 1 runs from 6 to 10, while set 2 is more spread out, from 4 to 12.

```
Set 1:          •  •  •  •  •
Set 2:    •     •     •     •     •
       ---+--+--+--+--+--+--+--+--+---
          4  5  6  7  8  9 10 11 12
```

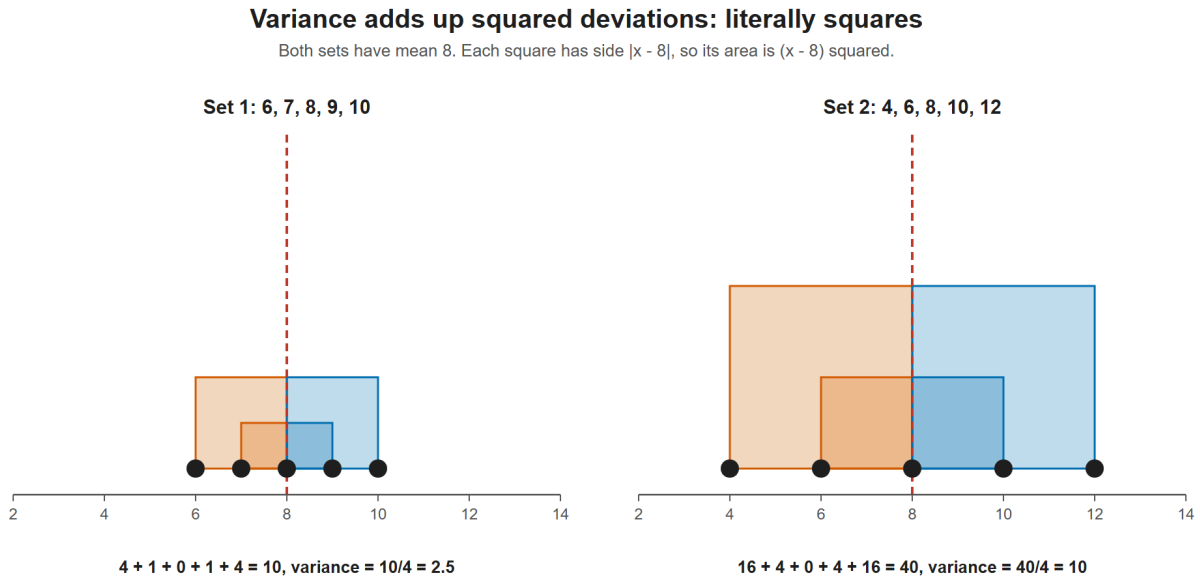
Set 1:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1} = \frac{2^2 + 1^2 + 0^2 + 1^2 + 2^2}{5 - 1} = \frac{10}{4} = 2.5$$

Set 2:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1} = \frac{4^2 + 2^2 + 0^2 + 2^2 + 4^2}{5 - 1} = \frac{40}{4} = 10$$

Set 2's variance is four times larger, because its points sit farther from the mean.



Each squared deviation is drawn as an actual square with side |x - 8|. Doubling the distances in Set 2 makes every square four times bigger, so the variance rises from 2.5 to 10.

Key takeaways

- Sample variance: square each deviation, add them, divide by $n - 1$.
- Variance = SD^2. Its units are the **square** of the data's units.
- Two data sets can have the **same mean** but **very different variances**.